TELCO CUSTOMER CHURN | RETENTION ANALYSIS

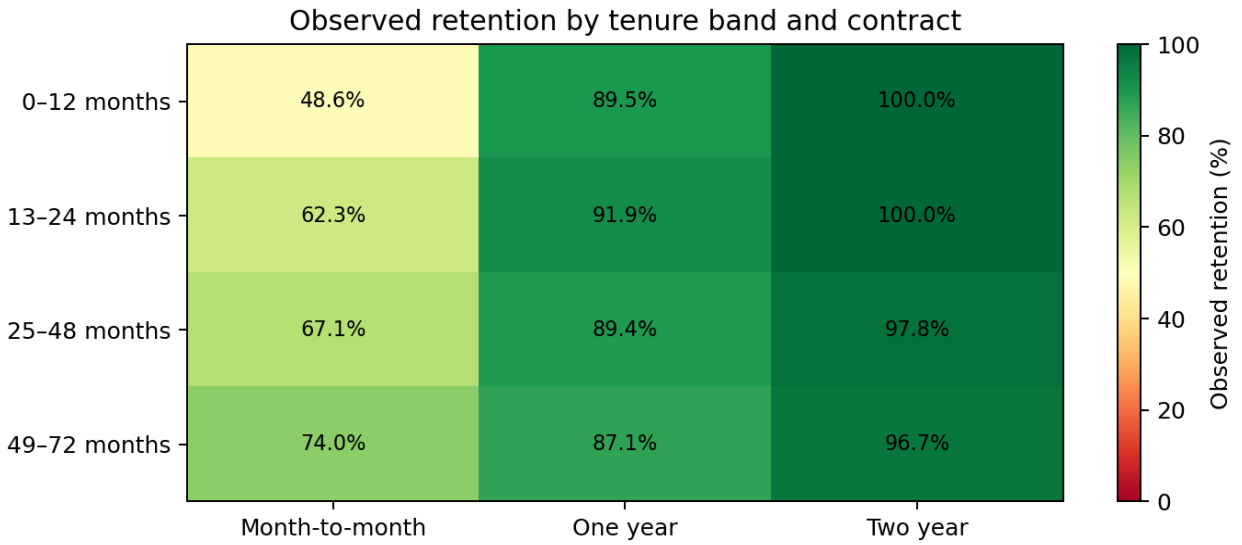
Datheon Internship • Task 3 • IBM Telco Customer Churn dataset

Executive summary

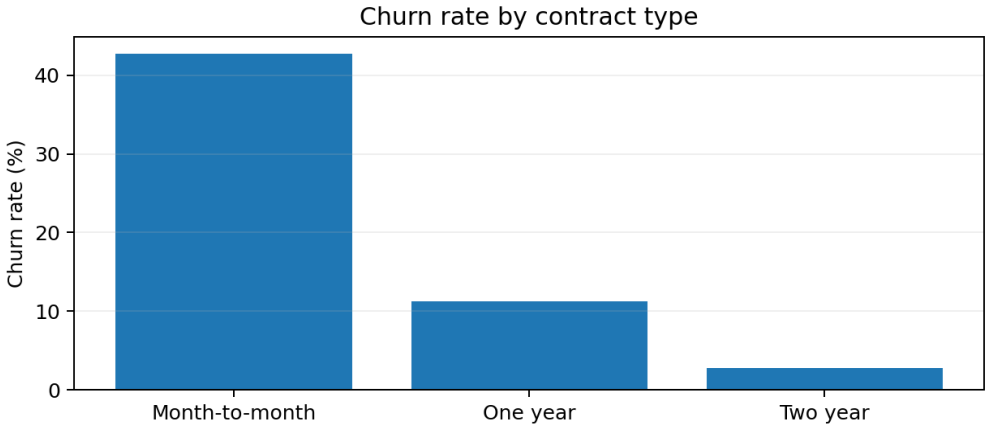
Total customers	Churned customers	Churn rate	Retention rate
7,043	1,869	26.54%	73.46%

Cohort-style retention matrix

The matrix reports the share of customers not marked as churned within each current-tenure band and contract type.

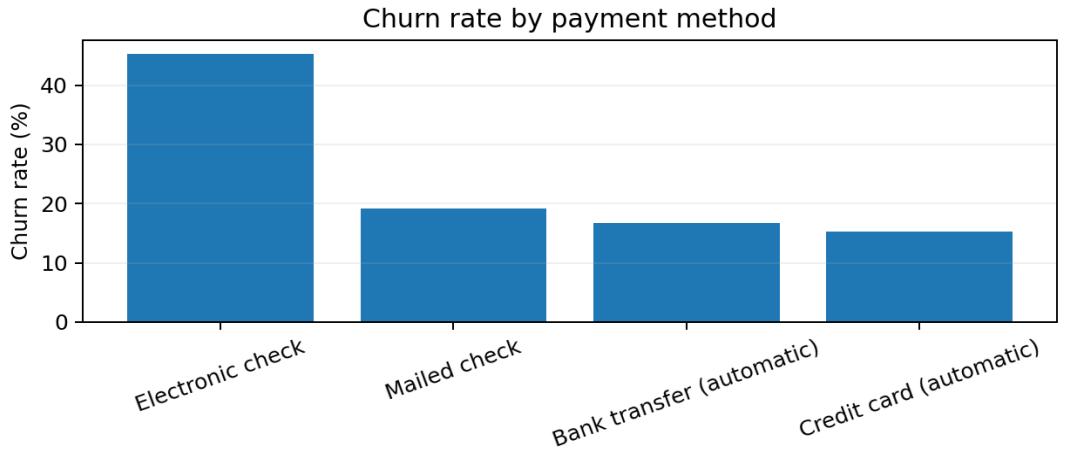


Churn drivers by contract and payment


Churn rate by contract type
Churn rate (%)
0
10
20
30
40
Month-to-month
One year
Two year

SEGMENT FINDINGS & RETENTION STRATEGY

Payment and service segmentation



Contract type

Segment	Customers	Churned	Churn rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

Payment method

Segment	Customers	Churned	Churn rate
Bank transfer (automatic)	1,544	258	16.71%
Credit card (automatic)	1,522	232	15.24%
Electronic check	2,365	1,071	45.29%
Mailed check	1,612	308	19.11%

Internet service

Segment	Customers	Churned	Churn rate

DSL	2,421	459	18.96%
Fiber optic	3,096	1,297	41.89%
No	1,526	113	7.40%

Key findings and recommendations

1. Overall churn: 26.54% of customers (1,869 of 7,043) have churned.

2. Contract: Month-to-month churn is 42.7%, compared with 2.8% for two-year contracts. Offer clear annual-plan incentives and early renewal outreach.

3. Payment: Electronic-check churn is 45.3%, the highest payment-method rate. Simplify automatic payment setup and billing education.

4. Internet service: Fiber-optic churn is 41.9%. Review service reliability, installation, pricing and support response.

5. Early tenure: Use 30/60/90-day onboarding and proactive service check-ins, especially for newer month-to-month customers.

6. Support: Test support/security bundles for relevant customer groups and measure outcomes through controlled experiments.

Data quality and limitations

The dataset contains 7,043 unique customer records. 11 TotalCharges values are blank and were retained as missing rather than imputed. No duplicate customer IDs were found. The dataset has tenure but no signup date or common observation date; consequently, the tenure-band matrix is a cross-sectional retention segmentation, not a true acquisition-vintage cohort or survival curve. Churn associations are descriptive and do not prove causation.